NOETHERIAN LOCAL K-ALGEBRAS WHOSE COMPLETED TENSOR IS A NONDEGENERATE QUADRIC POWER SERIES RING: THREE EQUIVALENT CHARACTERIZATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction of (3) to (1) via irreducibility of the quadric, the exclusion argument for dimension, the full lifting induction with the congruence modulo m^4 and the nondegeneracy basis step, the completeness limit, and the kernel identification with its degree contradiction are all retained verbatim.

Editorial note. The label lemma-nodal-algebraic names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Depth and dimension estimates on local rings. Algebra, Lemmas algebra-lemma-criterion-reduced, algebra-lemma-regular-graded, algebra-lemma-bound-depth. Rules out the Artinian and the dimension 2 (regular) cases in the implication (1) => (2), forcing depth(A) = 1.
- Lifting residue fields and completion generalities. More on Algebra, Lemmas more-algebra-lemma-completion-depth and more-algebra-lemma-lift-residue-field; Algebra, Lemma algebra-lemma-completion-generalities. Lets one assume A complete with a coefficient field kappa, so that the successive improvement of generators converges.

Source reference. Original lemma 00LK: sources/stacks-chow/tags.txt, source lines 790–790. Prerequisite pointer invoked at proof line(s) 4123; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0315: sources/stacks-chow/tags.txt, source lines 3972–3972. Prerequisite pointer invoked at proof line(s) 4171; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 031R: sources/stacks-chow/tags.txt, source lines 3994–3994. Prerequisite pointer invoked at proof line(s) 4118; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00NO: sources/stacks-chow/tags.txt, source lines 866–866. Prerequisite pointer invoked at proof line(s) 4121; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C48: sources/stacks-curves/pinned-original-source.tex, source lines 4034–4034. Prerequisite pointer invoked at proof line(s) 4172; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07NW: sources/stacks-chow/tags.txt, source lines 9776–9776. Prerequisite pointer invoked at proof line(s) 4128; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C34: sources/stacks-chow/tags.txt, source lines 15179–15179. Prerequisite pointer invoked at proof line(s) 4131; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian local k-algebra. The following are equivalent*

1. *κ/k is separable, A is reduced, $\dim_\kappa(\mathfrak{m}/\mathfrak{m}^2) = 2$, and there exists a nondegenerate $q \in Sym^2_\kappa(\mathfrak{m}/\mathfrak{m}^2)$ which maps to zero in $\mathfrak{m}^2/\mathfrak{m}^3$,*
2. *κ/k is separable, depth$(A) = 1$, $\dim_\kappa(\mathfrak{m}/\mathfrak{m}^2) = 2$, and there exists a nondegenerate $q \in Sym^2_\kappa(\mathfrak{m}/\mathfrak{m}^2)$ which maps to zero in $\mathfrak{m}^2/\mathfrak{m}^3$,*
3. *κ/k is separable, $A^\wedge \cong \kappa[[x, y]]/(ax^2 + bxy + cy^2)$ as a k-algebra where $ax^2 + bxy + cy^2$ is a nondegenerate quadratic form over κ.*

Proof. Assume (3). Then $A^\wedge$ is reduced because $ax^2+bxy+cy^2$ is either irreducible or a product of two nonassociated prime elements. Hence $A \subset A^\wedge$ is reduced. It follows that (1) is true.

Assume (1). Then A cannot be Artinian, since it would not be reduced because $\mathfrak{m} \neq (0)$. Hence $\dim(A) \geq 1$, hence $\text{depth}(A) \geq 1$ by Algebra, Lemma 031R. On the other hand $\dim(A) = 2$ implies A is regular which contradicts the existence of q by Algebra, Lemma 00NO. Thus $\dim(A) \leq 1$ and we conclude $\text{depth}(A) = 1$ by Algebra, Lemma 00LK. It follows that (2) is true.

Assume (2). Since the depth of A is the same as the depth of $A^\wedge$ (More on Algebra, Lemma 07NW) and since the other conditions are insensitive to completion, we may assume that A is complete. Choose $\kappa \to A$ as in More on Algebra, Lemma 0C34. Since $\dim_\kappa(\mathfrak{m}/\mathfrak{m}^2) = 2$ we can choose $x_0, y_0 \in \mathfrak{m}$ which map to a basis. We obtain a continuous κ-algebra map

$$\kappa[[x, y]] \longrightarrow A$$

by the rules $x \mapsto x_0$ and $y \mapsto y_0$. Let q be the class of $ax_0^2 + bx_0y_0 + cy_0^2$ in $\text{Sym}^2_\kappa(\mathfrak{m}/\mathfrak{m}^2)$. Write $Q(x, y) = ax^2 + bxy + cy^2$ viewed as a polynomial in two variables. Then we see that

$$Q(x_0, y_0) = ax_0^2 + bx_0y_0 + cy_0^2 = \sum\nolimits_{i+j=3} a_{ij}x_0^i y_0^j$$

for some a_{ij} in A. We want to prove that we can increase the order of vanishing by changing our choice of x_0, y_0. Suppose that $x_1, y_1 \in \mathfrak{m}^2$. Then

$$Q(x_0 + x_1, y_0 + y_1) = Q(x_0, y_0) + (2ax_0 + by_0)x_1 + (bx_0 + 2cy_0)y_1 \bmod \mathfrak{m}^4$$

Nondegeneracy of Q means exactly that $2ax_0 + by_0$ and $bx_0 + 2cy_0$ are a κ-basis for $\mathfrak{m}/\mathfrak{m}^2$, see discussion preceding the lemma. Hence we can certainly choose

$x_1, y_1 \in \mathfrak{m}^2$ such that $Q(x_0 + x_1, y_0 + y_1) \in \mathfrak{m}^4$. Continuing in this fashion by induction we can find $x_i, y_i \in \mathfrak{m}^{i+1}$ such that

$$Q(x_0 + x_1 + \ldots + x_n, y_0 + y_1 + \ldots + y_n) \in \mathfrak{m}^{n+3}$$

Since A is complete we can set $x_\infty = \sum x_i$ and $y_\infty = \sum y_i$ and we can consider the map $\kappa[[x, y]] \longrightarrow A$ sending x to x_∞ and y to y_∞. This map induces a surjection $\kappa[[x, y]]/(Q) \longrightarrow A$ by Algebra, Lemma 0315. By Lemma 0C48 the kernel of $k[[x, y]] \to A$ is principal. But the kernel cannot contain a proper divisor of Q as such a divisor would have degree 1 in x, y and this would contradict $\dim(\mathfrak{m}/\mathfrak{m}^2) = 2$. Hence Q generates the kernel as desired. □